

2025 – 2026 Biomolecular Engineering and Bioinformatics: Bioinformatics

Math & Statistics •MATH 3 or Math Placement MATH 19A Calculus I [F/W/Sp] •MATH 19A MATH 19B Calculus II [F/W/Sp] •MATH 3 or Math Placement AM 10 or Math 21 Mathematical Methods for Engineers I [F/W/Sp] •MATH 19A, or Math 19B; prior programming recommended CSE 16 Discrete Math [F/W/Sp] •MATH 19B (AM 30 or Math 23B or Math 22 recommended) STAT 131 Intro to Probability Theory [F/W/Sp] •STAT 131 STAT 132 Classical and Bayesian Inference [F/W/Sp] OR •Math 19B and CSE 30 CSE 40 Machine Learning Basics [F/W/Sp] •Math 19B and AM 10 AM 30 OR Math 23A Multivariate Calculus for Engineers & CSE 142 Machine Learning OR •CSE 40 or STAT 132; CSE 101; AM 30 or Math 23A; and STAT 131 CSE 144 Applied Machine Learning	Chemistry Select one of the following General Chemistry series: <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •Math 2 or math placement CHEM 3A General Chemistry [F/W/Sp] </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •Chem 3A CHEM 3B/BL General Chemistry [F/W/Sp] </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •Chem 3B CHEM 3C/CL General Chemistry [F/W/Sp] </td> </tr> </table> OR <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •Chem 4 Prep ALEKS module. Math 3 or math placement. CHEM 4A/AL Advanced General Chemistry Not offered 2025-26 </td> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •Chem 4A CHEM 4B/BL Advanced General Chemistry Not offered 2025-26 </td> </tr> </table>	•Math 2 or math placement CHEM 3A General Chemistry [F/W/Sp]	•Chem 3A CHEM 3B/BL General Chemistry [F/W/Sp]	•Chem 3B CHEM 3C/CL General Chemistry [F/W/Sp]	•Chem 4 Prep ALEKS module. Math 3 or math placement. CHEM 4A/AL Advanced General Chemistry Not offered 2025-26	•Chem 4A CHEM 4B/BL Advanced General Chemistry Not offered 2025-26	Biology, Organic Chemistry, and Biochemistry <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •CHEM 3A, or CHEM 4A BIOL 20A Cell and Molecular Biology [F/W/Sp] </td> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •CHEM 3C or CHEM 4B CHEM 8A Organic Chemistry [F/W] </td> </tr> </table> <table style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <tr> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •BIOL 20A BME 105 (Strongly Recommended) Genetics in the Genomics Era [W/Sp] </td> <td style="width: 10%; text-align: center; vertical-align: middle;">OR</td> <td style="width: 40%; border: 1px solid black; padding: 5px; text-align: center;"> •BIOL 20A and BIOE 20B BIOL 105 Genetics [F/W/Sp] </td> </tr> </table> Select one of the following BME 101 Molecular Biology for Biomolecular Engineers [F] OR •CHEM 8A CHEM 8B Organic Chemistry [W/Sp] AND one of the following <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •CHEM 8B CHEM 103 Biochemistry [Sp] </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •CHEM 8B and BIOL 20A BIOC 100A Biochemistry and Molecular Biology [F] </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •BIOL 20A and BIOE 20B; CHEM 8B or CHEM 112A BIOL 100 Biochemistry [F/W/Sp] </td> </tr> </table>	•CHEM 3A, or CHEM 4A BIOL 20A Cell and Molecular Biology [F/W/Sp]	•CHEM 3C or CHEM 4B CHEM 8A Organic Chemistry [F/W]	•BIOL 20A BME 105 (Strongly Recommended) Genetics in the Genomics Era [W/Sp]	OR	•BIOL 20A and BIOE 20B BIOL 105 Genetics [F/W/Sp]	•CHEM 8B CHEM 103 Biochemistry [Sp]	•CHEM 8B and BIOL 20A BIOC 100A Biochemistry and Molecular Biology [F]	•BIOL 20A and BIOE 20B; CHEM 8B or CHEM 112A BIOL 100 Biochemistry [F/W/Sp]
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Bioinformatics & Elective <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> BME 80G Bioethics in the 21st Century [Sp] </td> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •BME 105 or BIOL 105 or BIOC 100A or BIOL 100 or CHEM 103 or declared BMEB major BME 110 Computational Biology Tools [F/W/Sp] </td> </tr> </table> <table style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <tr> <td style="width: 50%; border: 1px solid black; padding: 5px; text-align: center;"> •ELWR and BIOL 20A BME 185 Technical Writing for Biomolecular Engineers [F/W/Sp] </td> <td style="width: 50%;"></td> </tr> </table> Elective <i>One of the following (course used as an Elective cannot be used to satisfy other major requirements):</i> AM 115, AM 147, BME 118, BME 122H, BME 128, BME 128L, BME 130, BME 132, BME 140, BME 175, BME 177, BME 177L, BME 178, BIOC 100B, CSE 142, CSE 144, CSE 182, METX 100, METX 140, or 5-unit BME grad course (BME 201-279).	BME 80G Bioethics in the 21st Century [Sp]	•BME 105 or BIOL 105 or BIOC 100A or BIOL 100 or CHEM 103 or declared BMEB major BME 110 Computational Biology Tools [F/W/Sp]	•ELWR and BIOL 20A BME 185 Technical Writing for Biomolecular Engineers [F/W/Sp]		Programming <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •BIOL 20A BME 160 (6 units)² Research Programming in the Life Sciences [F/W] </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •BME 160 or BME 205 BME 163 Applied Visualization and Analysis of Scientific Data [Sp] </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •CSE 20 or BME 160; and Math 3 or Math 19A CSE 30 (7 units) Programming Abstractions: Python [F/W/Sp] </td> </tr> </table> <table style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <tr> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •CSE 16, CSE 20 or BME 160, CSE 30 and Math 19B CSE 101P Introduction to Data Structures and Algorithms in Python [Sp] </td> <td style="width: 33%; text-align: center; vertical-align: middle;">OR</td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •CSE 12 or BME 160 CSE 13S (7 units) Computer Systems and C Programming [F/W/Sp] </td> </tr> </table> <table style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <tr> <td style="width: 50%;"></td> <td style="width: 10%; text-align: center; vertical-align: middle;">AND</td> <td style="width: 40%; border: 1px solid black; padding: 5px; text-align: center;"> •CSE 12 or BME 160 and 13S and CSE 16 and CSE 30 and Math 19B CSE 101 Introduction to Data Structures and Algorithms [F/W/Sp] </td> </tr> </table>	•BIOL 20A BME 160 (6 units)² Research Programming in the Life Sciences [F/W]	•BME 160 or BME 205 BME 163 Applied Visualization and Analysis of Scientific Data [Sp]	•CSE 20 or BME 160; and Math 3 or Math 19A CSE 30 (7 units) Programming Abstractions: Python [F/W/Sp]	•CSE 16, CSE 20 or BME 160, CSE 30 and Math 19B CSE 101P Introduction to Data Structures and Algorithms in Python [Sp]	OR	•CSE 12 or BME 160 CSE 13S (7 units) Computer Systems and C Programming [F/W/Sp]		AND	•CSE 12 or BME 160 and 13S and CSE 16 and CSE 30 and Math 19B CSE 101 Introduction to Data Structures and Algorithms [F/W/Sp]	
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Bioinformatics Capstone Choose one sequence <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •BME 160 or CSE 30; STAT 131 or CSE 107; prev. or concurrent enrollment in Biochemistry BME 205 Bioinformatics Models and Algorithms </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •BME 205 BME 230A Introduction to Computational Genomics and Systems Biology </td> <td style="width: 33%; border: 1px solid black; padding: 5px; text-align: center;"> •BME 129B or BME 230A BME 129C Project Design and Implementation in Biomolecular Engineering III </td> </tr> </table> BME 195 Senior Thesis (15 units over 3 quarters) [F/W/Sp]			•BME 160 or CSE 30; STAT 131 or CSE 107; prev. or concurrent enrollment in Biochemistry BME 205 Bioinformatics Models and Algorithms	•BME 205 BME 230A Introduction to Computational Genomics and Systems Biology	•BME 129B or BME 230A BME 129C Project Design and Implementation in Biomolecular Engineering III										
•BME 160 or CSE 30; STAT 131 or CSE 107; prev. or concurrent enrollment in Biochemistry BME 205 Bioinformatics Models and Algorithms	•BME 205 BME 230A Introduction to Computational Genomics and Systems Biology	•BME 129B or BME 230A BME 129C Project Design and Implementation in Biomolecular Engineering III													

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Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Exit Requirements

Students are required to submit a portfolio, exit survey, and attend an exit interview. The portfolios must be turned in by the last day of the quarter of graduation, and will be reviewed quarterly by the undergraduate director. Exit interviews are scheduled during the last week of the quarter by Baskin Engineering advising office, generally as small group interviews. Additional information can be found in the program catalog statement.

1. Portfolio
2. Exit Survey
3. Exit Interview

Student name: